

Name:

ID:

A segment of size **3584 bytes**, including **48 bytes** of header was fragmented for transmitting in a link with MTU=**X bytes**. The **9th** packet has its MF bit reset. The last fragment's data size is given as (**X- network layer header+2**)

- I. Find out the total number of fragmented packets. [2]
- II. Calculate the value of X. [7]
- III. If the fragment offset of last segment and second last segment is M and N,
Is the $M/N = (X - \text{network layer header})$ statement true?
Provide a detailed explanation. [6]